

# Mainul Islam

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## Education

### B.Sc. in Electrical and Electronic Engineering

[Islamic University of Technology \(IUT\)](#), Gazipur, Bangladesh

Jan 2020 – Jun 2024

CGPA: 3.87 / 4.00 (4.00/4.00 in last year)

#### Thesis -

- MonoSIM, an open-source SIL platform developed for testing autonomous control algorithms on Ackermann vehicles.
- The system was designed with a monocular camera vision system with a sliding window based lane detection method.
- The system showed that PID had lower tracking error while MPC gave smoother, constraint-aware control.

## Publications

### Peer-Reviewed Journal Articles

- A. U. R. Adib, **M. Islam**, M. S. Abid, R. Ahshan, “A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI,” *Solar Energy*, vol. 302, 2025. (Impact Factor: 6.6) DOI: [10.1016/j.solener.2025.114058](https://doi.org/10.1016/j.solener.2025.114058)
- S. Rahman, A. Banik, M. K. Hossain, **M. Islam**, A. Al-Hysam, I. Ahmed, “Topological survey of zeta power converters from classical designs to emerging architectures,” *Energy Conversion and Management: X*, art. no. 102222, 2026. (Impact Factor: 8.8) DOI: [10.1016/j.ecmx.2026.102222](https://doi.org/10.1016/j.ecmx.2026.102222)

### Peer-Reviewed Conference Papers

- S. Rahman, N. Hasin, **M. Islam**, M. Z. A. Rony, G. Sarowar, “MonoSIM: An open-source SIL framework for Ackermann vehicular systems with monocular vision,” *IEEE 12th Intl. Conf. on Automation, Robotics and Applications (ICARA)*, 2026. DOI: [10.1109/ICARA69401.2026.11480327](https://doi.org/10.1109/ICARA69401.2026.11480327)
- N. Hasin, M. A. R. Nishat, **M. Islam**, K. S. A. Hasan, A. Newaz, “A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR,” *IEEE Intl. Conf. on AI and Data Analytics (ICAD)*, 2026. DOI: [10.1109/ICAD69378.2026.11608728](https://doi.org/10.1109/ICAD69378.2026.11608728)
- H. Haque, **M. Islam**, M. A. I. Alvy, M. S. Islam, M. A. Razzak, “Long-term energy demand analysis using machine learning algorithms: A case study in Bangladesh,” *IEEE ICECIE*, 2024. DOI: [10.1109/ICECIE63774.2024.10815643](https://doi.org/10.1109/ICECIE63774.2024.10815643)
- M. S. Tanvir, F. A. Tonmoy, M. M. Islam, **M. Islam**, M. Hasan, “Optimizing master-slave broadcast and unicast communication in multi-node STM32 networks using UART and RS485 protocols,” *IEEE ICCIT*, 2024. DOI: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)

## Professional Experience

**Lecturer (Part-Time), Dept. of EEE** — [Ahsanullah University of Science and Technology](#)

Jul 2025 – Jun 2026

- Conducted C++ Programming and Digital Signal Processing courses.
- Delivered lectures, prepared laboratory manuals, and conducted student assessment.

**R&D Engineer (Robotics & Embedded Systems)** — [Cybernetics Hi-Tech Solutions](#)

Oct 2024 – Present

Delivery of robotic and embedded systems from requirements through field commissioning, spanning control electronics, firmware, autonomy software, and operator interfaces.

#### [Man-Transportable EOD Robot \(Jontro Soinik 2.0\)](#)

- Developed the motion-control library for eleven servo drives, implementing the CANopen (CiA-402) standard.
- Contributed to the arm's inverse kinematics, integrated with MoveIt 2 for collision checking.
- Wrote the STM32 firmware carrying the operator's control signal over an Ethernet link via optical fiber.
- Developed the operator application, low-latency video feeds and a 3D model mirroring the robot's actual posture.

### Autonomous Guided Vehicle — CYBERFLEET

- Designed the complete ROS 2 system integration for a three-vehicle warehouse robot fleet, covering how each vehicle establishes its position, plans a route, follows it, and coordinates with the others.
- Implemented an extended Kalman filter combining wheel-encoder motion with two cameras observing Apriltags.
- Built the marker-based localisation pipeline: camera calibration and pose estimation from each observed marker.
- Contributed to the route planning that treats time as a planning dimension, preventing both collisions and deadlocks.
- Implemented LiDAR-based obstacle detection.

### Autonomous Target-Tracking Drone

- Contributed to the implementation of live onboard target detection and tracking, running on a Pi 5 with an AI Hat.
- Created the aerial training dataset and evaluated alternative tracking methods.

### Toll-Booth Automation

- Developed the lane-controller firmware on an STM32 microcontroller, structured as a table-driven state machine.
- Integrated inductive loop coils, an infrared exit sensor, and licence-plate recognition to detect vehicles.
- Designed three custom circuit boards with optical isolation, surge and reverse-polarity protection.

### **Power Systems Design Lead — IUT Mars Rover — PROJECT ALTAIR**

Jan 2023 – Jun 2024

- Simulated and analysed custom circuits and designed PCBs up to fabrication.
- Built custom battery management systems (BMS) compatible with a wide range of cell arrangements.

## **Undergraduate Projects**

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### SAP-1 8-Bit Computer Design

- Simulated an 8-bit computer in Proteus, redesigning it to cut the instruction cycle from 6 T-states to 4.

### Pulse Width Modulation by 8051

- Bare-metal 8051 firmware developed to measure the frequency of two signals and generate a summed-frequency output.

### Multi-Cell Analog BMS

- Software and hardware tested under-voltage protection for 2–6s cell configurations using logic gates only, with no microcontroller required.

## **Technical Skills**

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**Programming** : Python, C/C++, Embedded C, MATLAB, assembly.  
**Robotics and Control** : ROS 2, sensor fusion.  
**Embedded Systems** : STM32 platforms, CAN, Modbus.  
**Machine Learning** : PyTorch, TensorFlow.  
**Design Tools** : EAGLE, KiCAD, Easy eda, Proteus.

## **Honours and Achievements**

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- **International Rover Challenge 2024:** 6th globally & 1st in BD ([Project Altair](#)); ABEX Best Science Team.
- **European Rover Challenge 2023:** 16th globally ([Project Altair](#)).
- [MATLAB Cody](#): Global rank 117, with 914 problems solved and 26 badges.

## **References**

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### Prof. Dr. Golam Sarowar

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Professor, Department of EEE, (IUT)  
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### Rashed Mujib Noman

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Cybernetics Hi-Tech Solutions, Dhaka  
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